Christian Málaga-Chuquitaype

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RESEARCH PROFILE

I lead the **Emerging Structural Technologies** research group at Imperial College London, where my central question is whether the dynamic behaviour of structures can be designed from dynamical first principles — expressed in compact, general forms that hold across scales, configurations, and loading conditions — rather than merely resisted or described empirically after the fact.

That pursuit has produced foundational work on inerter-rocking systems including original Lagrangian formulations for non-smooth clutched dynamics, enabling design and optimization from first principles and confirmed at full structural scale, alongside advanced modelling and design in timber, and emerging programmes on extraterrestrial structures and seismic metamaterials. With 75+ peer-reviewed journal papers (several carrying highly-cited certificates), the Stanford/Elsevier Top 2% Scientist ranking since 2023, and £2M in competitive research income from UKRI, EU, British Council, and bilateral partnerships spanning Latin America, Turkey and Africa, my programme reflects both scientific depth and a long-standing commitment to developing structural engineering capacity in underserved communities worldwide.

Beyond Imperial, I serve as Associate Editor for three major scientific journals, a member of several leading editorial boards, the UK Deputy National Delegate to both the International Association for Earthquake Engineering and the European Association for Earthquake Engineering, and a member of ASCE, BSI and ISO committees drafting codified guidelines for structural design.

KEY METRICS

75+	130+	12	£2M+
Journal Papers	**Total Publications**	**PhDs Graduated**	**Research Income**
incl. 6 highly-cited	*journals, conf. & chapters*	*+ 11 active, 4 post-docs*	*UK · EU · LatAm · Africa*

RESEARCH INTERESTS

- Structural dynamics & non-smooth mechanics: rocking, clutching inerters, mixed Lagrangian formalism
- Performance-based design & assessment under extreme multi-hazard scenarios: seismic, blast, off-Earth
- AI-aided engineering: physics-informed machine learning, video- and sensor-based structural monitoring
- Seismic metamaterials and engineered periodicity for wave attenuation
- Computational modelling: damage-plasticity constitutive laws, long-term and life-cycle structural response
- Sustainable & low-cost seismic-resistant systems: cross-laminated timber, traditional vernacular technologies
- Experimental structural mechanics: large-scale testing, hybrid simulation, low-cost open instrumentation

EDUCATION

PhD in Structural Engineering *Imperial College London, UK · 2012*
Thesis: Seismic design and assessment of steel structures incorporating tubular members. Funded by the EPSRC and Tata Steel (Dorothy Hodgkin Postgraduate Award). Unwin Prize for Best PhD Thesis, Department of Civil & Environmental Engineering.

MSc in Earthquake Engineering *ROSE School, Universita degli Studi di Pavia (Italy) · 2007*
MProf MProf in Géodynamique, Exploration et Risques *Université Joseph Fourier (France) · 2007*
First term at Imperial College (all A* and A grades). Rose School A* GPA, Co-awarded. Full scholarship from the European Union.

BEng (Bachiller) in Civil Engineering *Universidad Nacional San Agustín de Arequipa, Peru · 2000*
Public Commendation (highest honour) for Best Degree Dissertation.

SELECTED HONOURS & AWARDS

2026 JSPS Long Term Invitational Fellow (Japan Society for the Promotion of Science)
2023– Stanford / Elsevier Global Top 2% Scientist Ranking
2025 Featured Paper Award, Engineering Structures, Elsevier [J68]
2025 Top Cited Paper, Earthquake Engineering & Structural Dynamics [J63]
2022 Top 10 Most Downloaded Paper, Earthquake Engineering & Structural Dynamics [J30]
2021 Top 10 Most Downloaded Paper, Earthquake Engineering & Structural Dynamics [J23]
2019 Best Research Paper, Institution of Structural Engineers UK [J21]
2018 Commendation for Excellence in Engineering Education, IStructE UK
2015 TK Hsieh Research Award — Best Paper, Institution of Civil Engineers
2012 Unwin Prize — Best PhD Thesis, Dept. of Civil & Environmental Engineering, Imperial College London
2007–11 Dorothy Hodgkin Postgraduate Award (EPSRC & Tata Steel)
2010 Prize for Highly Commended Poster (×2): Graduate School & CEE Dept, Imperial College
2009 Paviors Laing Award, Worshipful Company of Paviors (visit to UC Berkeley, USA)
2008 Huixian Earthquake Engineering Foundation Young Researcher Grant, 14th WCEE Beijing

ACADEMIC EMPLOYMENT

Imperial College London — Dept. of Civil & Environmental Engineering *2014 – present*
Lecturer (2014) → Senior Lecturer → Associate Professor → Reader in Structural Dynamics

- Director, Undergraduate Admissions (2022–present)
- Director, MSc Programme in Earthquake Engineering (2024–present)
- Post-doc and Fellows Champion (2019–present)
- Visiting Professor, Universidad del Bío-Bío, Chile (2024–present)

Teaching (UG & PG): Structural Dynamics; Design of Timber Structures; Seismic Design of Steel Structures; Building Resilient Structures: The Science and Art of Earthquake Engineering. Teaching (CPD): Structural component, Integrated Space Science & Engineering programme; IStructE Timber Design.

University of Tokyo — Dept. of Architecture — Visiting Researcher *2026*
AI-enabled video-based sensing with motion magnification for structural health monitoring.

Imperial College London — Research Associate *2012 – 2014*
ADBLAST (EU): large European project on industrial structures under vapour cloud explosions. Seismic response of composite timber/bamboo structures, in collaboration with Arup, for low-cost housing in El Salvador.

Universidad Católica de Santa María, Peru — Visiting Lecturer *2009*
Organised and delivered a postgraduate course in Earthquake Resistant Design.

Universidad Nacional San Agustín de Arequipa, Peru — Assistant Professor *2004 – 2005*
Undergraduate lecturing and management of the Faculty's social projects programme.

INDUSTRIAL & CONSULTING PRACTICE

Imperial Consultants *2013, 2015*
Expert opinion (for Advanta Global / Swiss Re) on causes of damage and collapse of high-profile structures.

Peruvian College of Engineers — Engineering Consultant *2004 – 2005*
Expert advice on structural safety of seismically damaged buildings following the 2001 Great Ica Earthquake.

PUBLICATIONS

Google Scholar · Scopus · ResearchGate · *Stanford/Elsevier Top 2% Scientist since 2023*

Selected Recent Journal Papers (75+ total; 6 highly-cited or most-downloaded certificates)

[J75] Zhang, Y., Lavan, O., Málaga-Chuquitaype, C. (2026). Efficient optimization of clutched inerter dampers using mixed Lagrangian formalism. *Computers & Structures, 322:108121*

[J74] Shi, D., Málaga-Chuquitaype, C. et al. (2026). Glubam roof trusses: Uncertainty quantification and partial safety factors calibration based on Bayesian methods. *Reliability Engineering & System Safety*

[J73] Chen, Y., Málaga-Chuquitaype, C., Gardner, L. (2025). Predicting local buckling in steel members under cyclic loads: A strain-based approach using the CSM. *Engineering Structures, 345A:121473*

[J68] Zahra, F., Málaga-Chuquitaype, C. (2025). Hazard-consistent performance comparison of ASCE and Eurocode-compliant steel moment frames. *Engineering Structures, 326:119492* **★ Editor's Feature Award**

[J66] Zhang, Y., Málaga-Chuquitaype, C., Lavan, O. (2024). Mixed Lagrangian Formulation for modelling structures with clutched inerter devices. *Earthquake Engineering & Structural Dynamics, 53(14):4203-4222*

[**J63]** Vicencio, F., Alexander, N.A., Málaga-Chuquitaype, C. (2024). Seismic structure-soil-structure interaction between inelastic structures. Earthquake Engineering & Structural Dynamics. 53(4):1446-1464 **★ Highly cited paper**

[J59] Shen, S., Málaga-Chuquitaype, C. (2024). Physics-informed artificial intelligence models for the seismic response prediction of rocking structures. *Data-Centric Engineering, 5:e110*

[J53] Maqdah, J., Memarzadeh, M., Kampas, G., Málaga-Chuquitaype, C. (2024). AI-aided exploration of lunar arch forms under in-plane seismic loading. *Acta Mechanica, 235:1517-1533*

[J47] Málaga-Chuquitaype, C. et al. (2022). Optimal arch forms under in-plane seismic loading in different gravitational environments. *Earthquake Engineering & Structural Dynamics, 51(6):1522-1539*

[J46] Málaga-Chuquitaype, C. (2022). Machine learning in structural design: an opinionated review. *Frontiers in Built Environment, 8*

[J30] Pan, X., Málaga-Chuquitaype, C. (2020). Seismic control of rocking structures via external resonators. *Earthquake Engineering & Structural Dynamics, 49:1180-1196* **★ Highly cited · Top 10 downloaded**

[J23] Thiers-Moggia, R., Málaga-Chuquitaype, C. (2019). Seismic protection of rocking structures with inerters. *Earthquake Engineering & Structural Dynamics, 48:528-547* **★ Highly cited · Top 10 downloaded**

[J21] Málaga-Chuquitaype, C., Ilkanaev, J. (2018). Novel digitally-manufactured wooden beams for floor vibration suppression. *Structures, 16:1-9* **★ Best Research Paper Award, IStructE**

[J10] Málaga-Chuquitaype, C., Elghazouli, A.Y. et al. (2014). Seismic behaviour of low-cost timber frames with cane and mortar walls. *Proc. ICE: Structures & Buildings, 167:693-703* **★ Best Research Paper, ICE**

Book Chapters

Málaga-Chuquitaype, C., Elghazouli, A.Y. (2016). Design of Timber Structures. *in Seismic Design of Buildings to Eurocode 8, Elghazouli Ed., Spon Press*

Zahra, F., Málaga-Chuquitaype, C. (2024). Seismic Behaviour of Damaged Steel Frames Retrofitted with Inerters. *in STESSA Proc., Mazzolani et al. Ed., Springer*

Selected Conference Papers (63+ refereed proceedings)

Earth and Space (2026) Organizer; EMI IC (2026); 18th European Conference on Earthquake Engineering (2026) 18th World Conference on Earthquake Engineering (2024) ×4; 3rd European Conference on Earthquake Engineering & Seismology (2022) ×6; World Conference on Timber Engineering (2020, 2018); EURODYN (2020, 2024); EMI Conference (2024) ×3; STESSA 2024; ICONHIC 2022 ×2.

RESEARCH FUNDING (SELECTED; TOTAL > £2M)

Period	Funder	Project / Title	Role	Total Value
2026–27	UKRI IAA	NOVA: Next generation of sustainable timber floors with vibration absorbing capabilities	PI	**£107k**
2026-28	H2020 (EU) MCSA	WAVES: Coupled dynamics and vibration control in integrated floating wind and wave energy structures	PI (Fellow: H Su)	**€260k**
2024–26	British Council	RESIBUILD: Resilient buildings innovated through ai-supported design and easy manufacturing	PI	**£100k**
2024-26	H2020 (EU)	ERIES-TRUST: Truly Resilient Timber Buildings	PI	**€90k**
2024–26	ANID	Proyecto ID 24I10158: Development of a timber-concrete modular construction system	Int. Co-I	**€211k**
2024	Imperial-MIT Fund	Towards sustainable infrastructure through structural optimization	PI	**£57k**
2021-23	UNSA Investiga	Low-cost sensing of critical facilities	Int. Co-I	**£900k**
2018	EPSRC GCRF	Improving seismic resilience in Latin America through low-cost quick sensing	PI	**£12k**
2016–19	TUBITAK	Seismic response of masonry structures with timber lacings	Int. Co-I	**£20k**

RESEARCH SUPERVISION

PhD Supervision — 12 Graduated · 11 Active · 4 Post-doctoral Researchers

Current PhD students (selected):

- Mohammed Ashour (2026–) — AI-enhanced video motion for infrastructure twining (KIST Scholarship)
- David Chalco Pari (2024–) — Hazard-consistent modelling of RC walls (Concytec Scholaship)
- Zilan Zhang (2024–) — Advanced computational modelling of seismic-resistant timber structures
- Atif Rasheed (2023–) — Low-cost seismic isolation (Commonwealth Scholaship) **★ SECED Fund, Paviors Laing Award, Global Gellows Fund, Best Dept. Poster**
- Yixuan Zhang (2023–) — Modelling and design of rocking structures (Dixon Scholarship)
- Shirley Shen (2022–) — Physics-informed AI for nonlinear seismic responses (Dixon Scholarship)
- Yu Cheng (2023–) — Nonlinear instabilities in metamaterials for vibration control (Skempton Scholarship)

Recent graduates (selected):

- Rodrigo Thiers (2020) — Seismic control of rocking structures with inerters **★ Unwin Prize — Best Thesis**
- Faridah Zahra (2024) — Hazard-consistent performance-based assessment of seel frames
- Carlos Melchor Placencia (2024) — Non-linear analysis of post-tensioned timber elements
- Eknara Junda (2024) — Machine learning for seismic response of timber structures
- Adeyanju Teslim-Balogun (2019) — Steel structures under blast and post-blast fire **★ Prize IStructE Poster**
- Leena Kibriya (2020); Zhan Lyu (2020); Luis Sirumbal (2019); Cagatay Demirci (2019); Vasileios Karagiannis (2017)

MSc/MEng: 75+ students supervised; multiple IStructE First Prize winners (Abeysekera 2014, Peper 2018), multiple Basu Prize winners (Aedo Maluje 2019, Majdad 2021), multiple Best Structures Project winners (Zhou 2025, Lee 2024, McLean 2020, Zu 2017)

SERVICE TO THE PROFESSION

Editorial Roles

- Associate Editor: Structures (Elsevier, 2024–); Innovative Infrastructure Solutions (Springer, 2022–); Engineering and Computational Mechanics (ICE, 2026–)
- Academic Editor: Shock and Vibration (Wiley, 2025–)
- Editorial Board: Engineering Structures (Elsevier, 2025–); Computers & Structures (Elsevier, 2025–); Bulleting of Earthquake Engineering (Springer, 2026–)
- Guest Editor: SI on AI-driven Structural Design & Optimisation, Engineering Structures (Elsevier, 2024–25)

Standards & Committees

- UK Deputy National Delegate, International Association of Earthquake Engineering (IAEE) & European Association of Earthquake Engineering (EAEE)
- Member, BSI B/525/5 (2024–); former Member BSI B/525/8 (2016–19); Member CEN/TC 250/SC 5/WG 11 (2024–); UK Representative ISO/TC 165/WG 10 (2025–)
- Member, ASCE Aerospace Division — Space Engineering & Construction TC: drafting LIEDAC Lunar Infrastructure Guidelines
- Invited Expert, CEN/TC250/WG11 drafting European Technical Specification for Numerically-Assisted Timber Design
- Elected Committee Member, SECED – UK Society for Earthquake and Civil Engineering Dynamics (2022–)

Review & External Examining

- Research council reviewer: UKRI Talent Panel, EPSRC Peer Review College (2022–), Swiss NSF (2018–), CONICYT Chile (2019–), CONCYTEC Peru (2018–)
- External examiner for PhD theses at: Oxford, UCL ×3, Bristol, ETH Zürich ×2, Edinburgh, Strathclyde, City x2, Griffith (AU), Galway (Ireland), University of Iceland (Iceland), Trieste ×2, Trento (Italy)
- Subject external examiner for degree programmes: Bristol (MSc Earthquake Engineering, 2022–25), Portsmouth, Aberdeen, Anglia Ruskin, Brighton

Outreach & Media

- Expert commentary: BBC, Associated Press / Globe and Mail, Times of India on the 2025 Myanmar earthquake and Bangkok high-rise collapse; BBC on 2017 Mexico City earthquake
- Television: Unearthed (TV Series, 2022); Disasters Engineered (TV Series, 2019)
- Knowledge transfer: short courses on Seismic Design of Steel Structures at Universidad Católica San Pablo and Universidad Nacional San Agustín (Peru)